

AutoEncoder on CelebA - Embedding Analysis Report

• Introduction

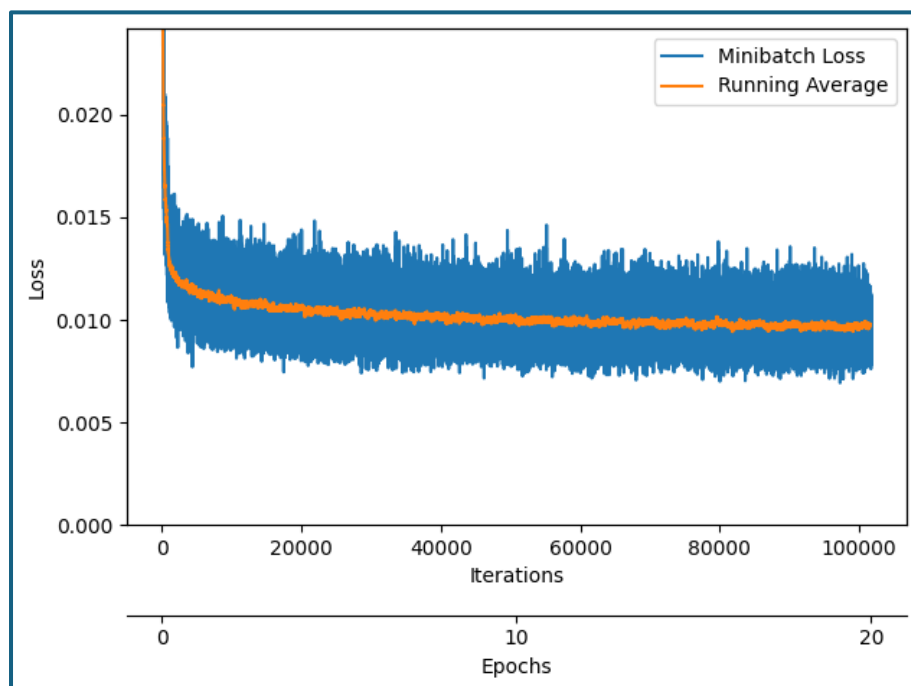
This report details using a Convolutional AutoEncoder on the CelebA dataset, encoding images into lower-dimensional spaces for reconstruction. Various embedding sizes (32, 64) were tested. PCA and t-SNE visualized the embeddings, with GIFs showing latent dimension manipulations.

• Tools and Techniques

- Dataset: CelebA (202,599 celebrity face images with 40 attribute annotations).
- Model: Convolutional AutoEncoder implemented in PyTorch
- Embedding Sizes: 32, 64.
- Visualization: PCA and t-SNE for latent space projection.
- Output: Loss plots, reconstructed images, latent space scatter plots, embedding histograms, and GIFs

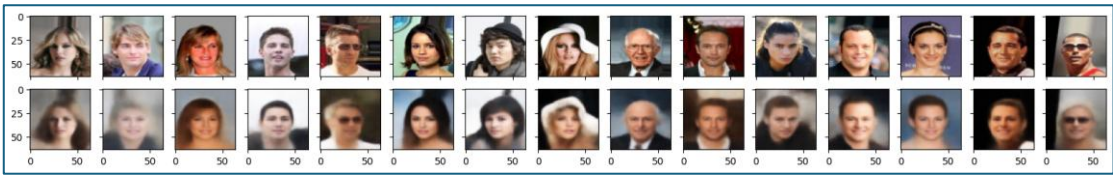
• Embedding Size = 32 ▪

• Training Loss

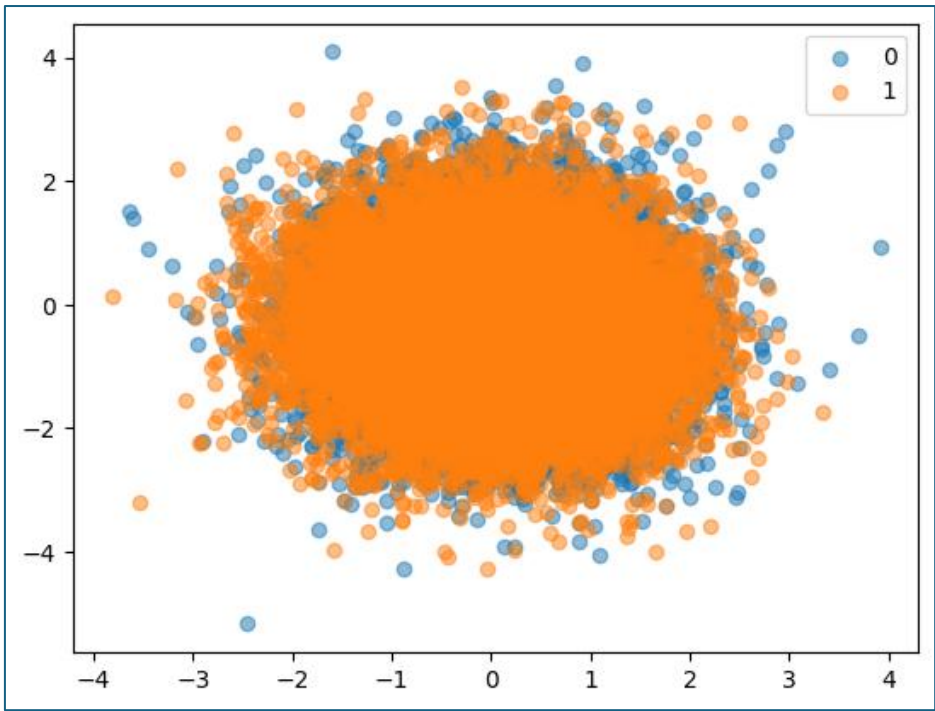


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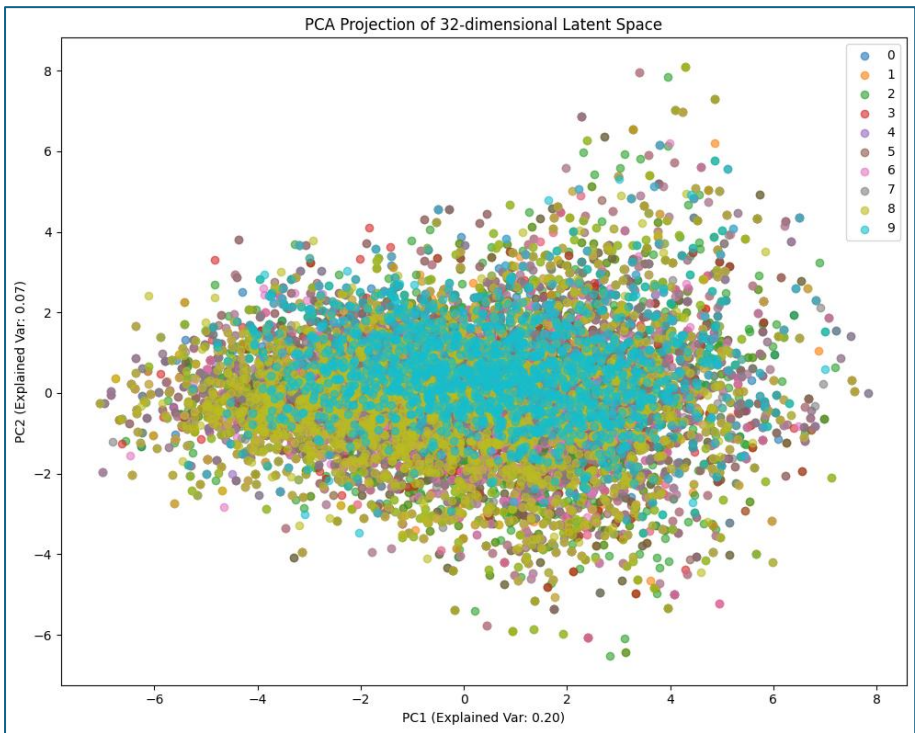
■ Reconstructed Images



■ latent space of the generated embeddings

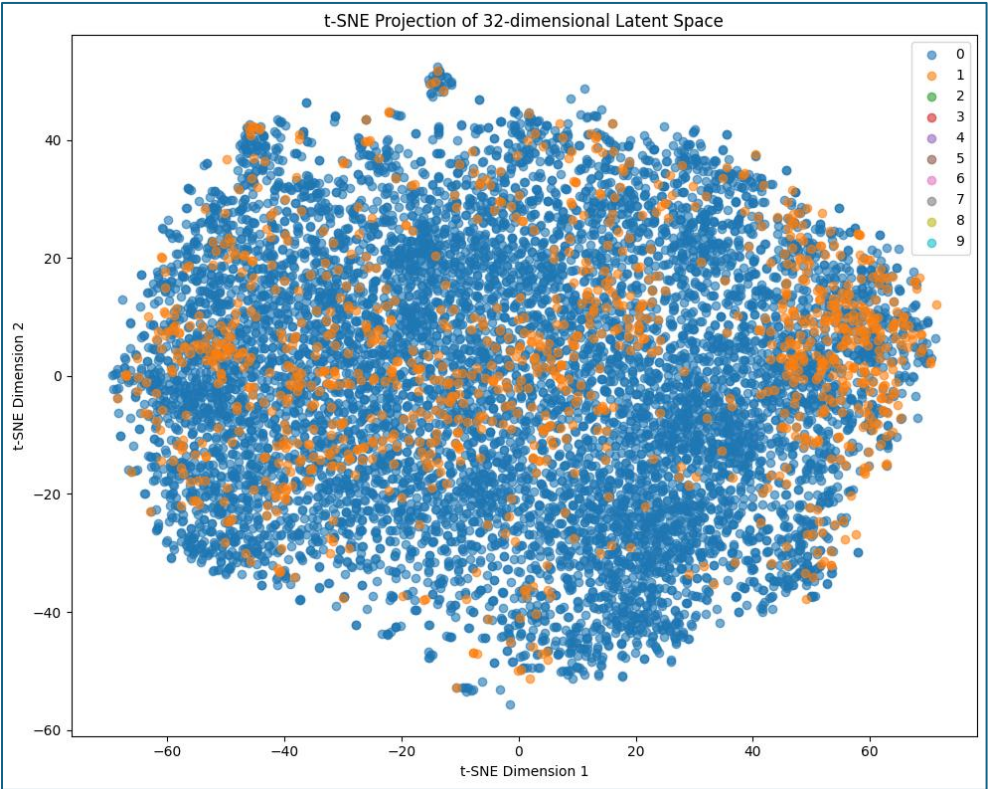


■ Latent Space Visualization (PCA)

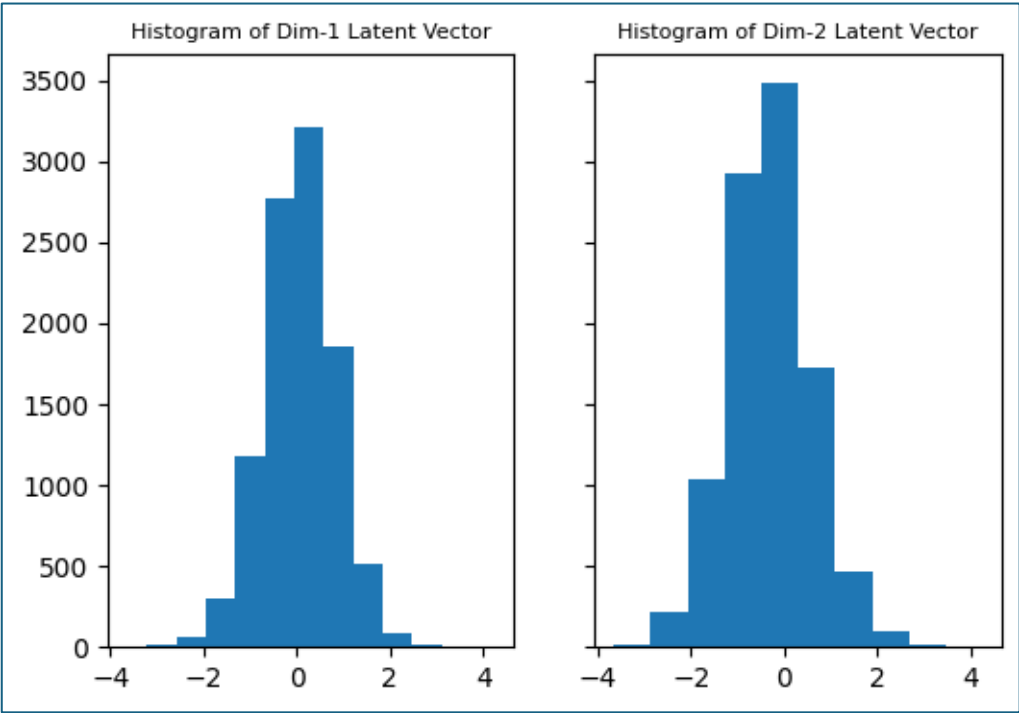


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Latent Space Visualization (t-SNE)

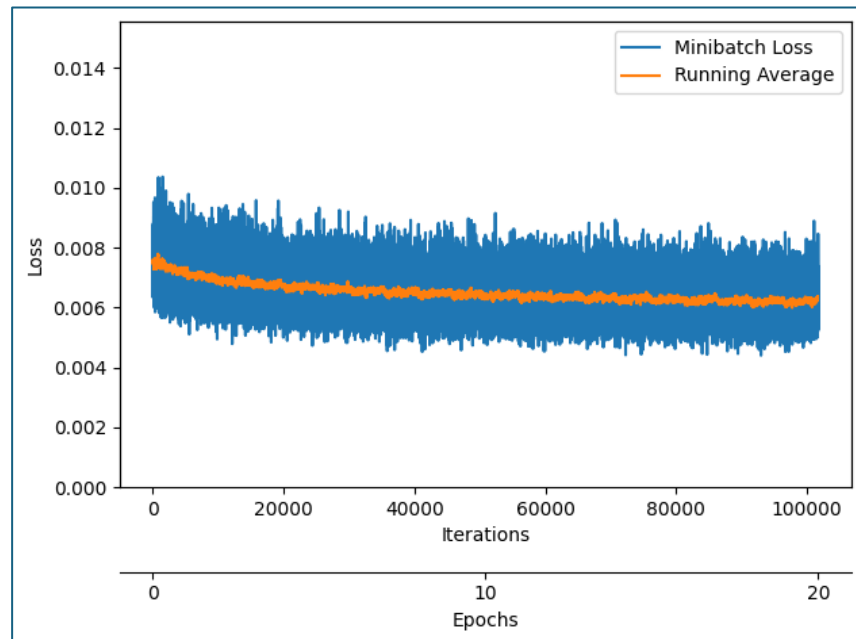


Embedding Histograms

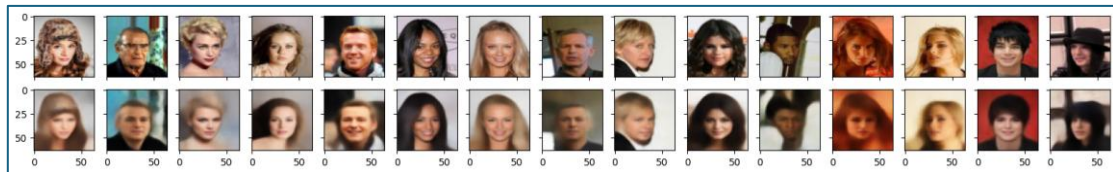


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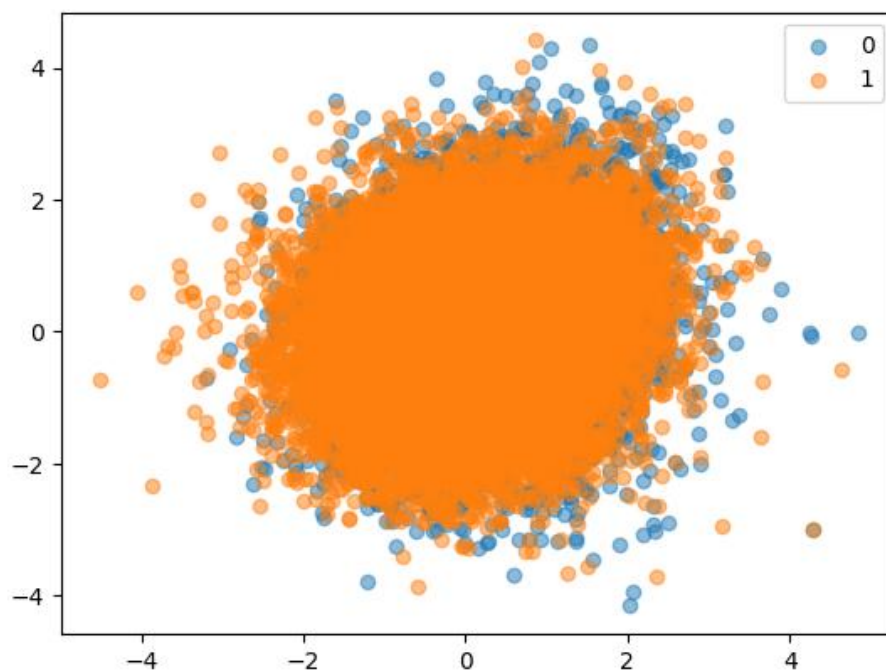
- Embedding Size = 64
- Training Loss



■ Reconstructed Images

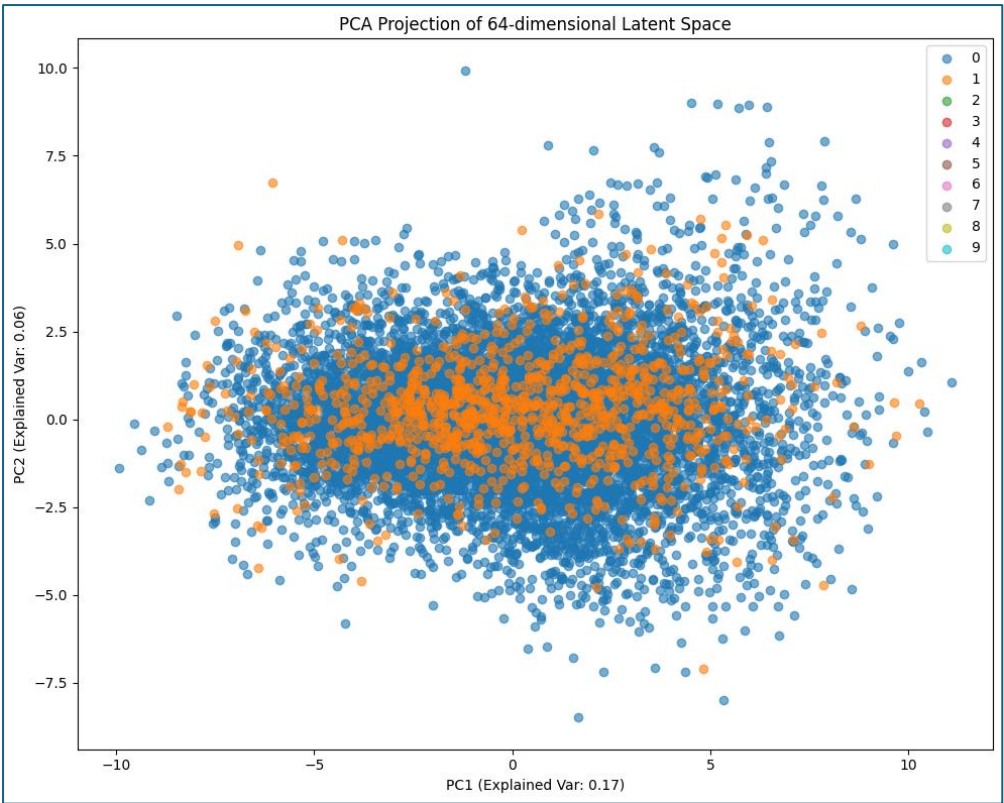


■ latent space of the generated embedding

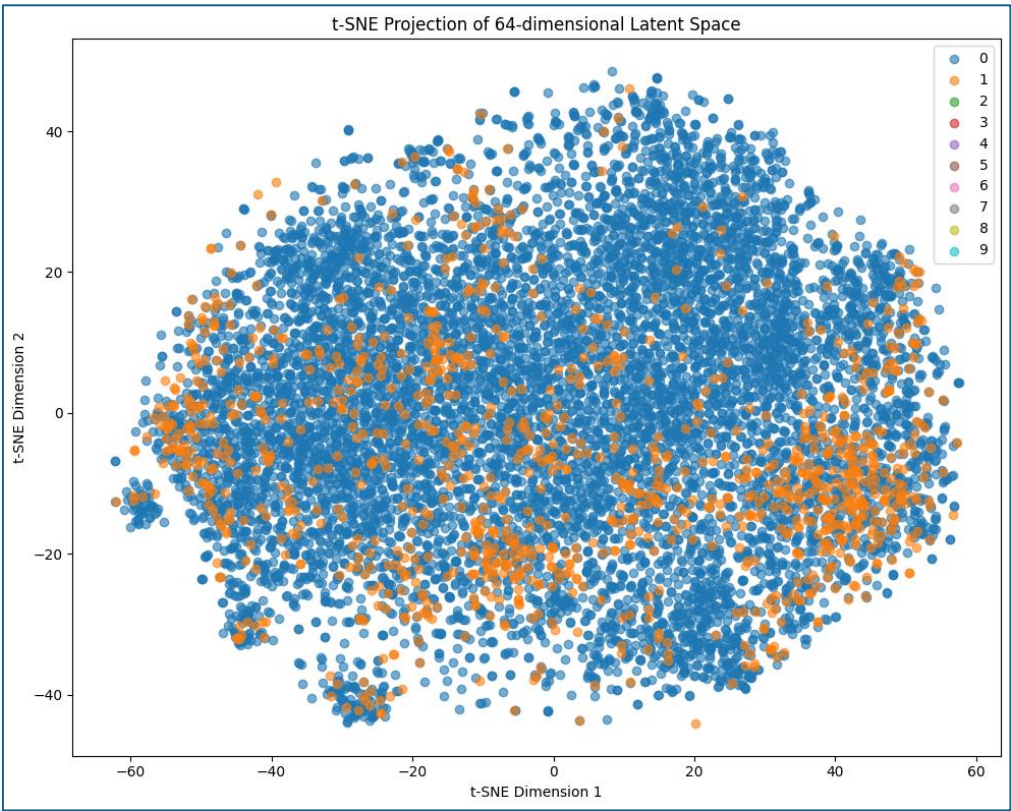


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■ Latent Space Visualization (PCA)

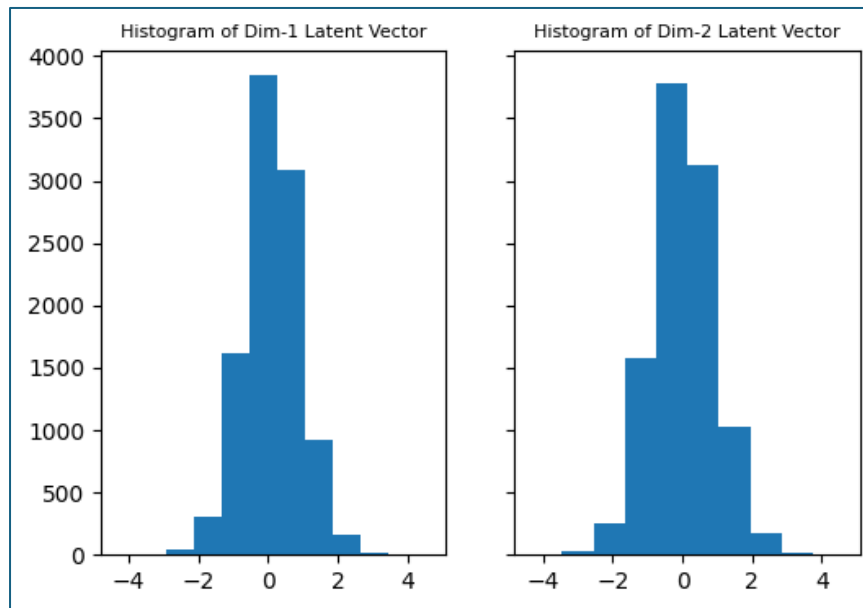


■ Latent Space Visualization (t-SNE)



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■ Embedding Histograms



• Latent Dimension Interpolation (Embedding Size = 32)

These animations show the reconstructed image as we vary one latent dimension at a time.

GIF - Varying Dimension 1



GIF - Varying Dimension 2



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- **Latent Dimension Interpolation (Embedding Size = 64)**

These animations show the reconstructed image as we vary one latent dimension at a time.

GIF - Varying Dimension 1



GIF - Varying Dimension 2

